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Surf-Snooping: USB crosstalk leakage attacks on wireless charging

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ABSTRACT

Various mobile devices such as smartphones, tablets, and computers have been increasingly incorporated with wireless charging technology. Nevertheless, this widespread adoption of wireless chargers has raised substantial concerns regarding privacy and security. The Universal Serial Bus (USB), serving as the primary power supply port, is widely acknowledged as a significant source of privacy vulnerabilities. This article introduces Surf-Snooping, a side-channel attack that is aimed at exploiting vulnerabilities in wireless charging systems. Through monitoring the voltage fluctuations of a nearby USB charging port, an attacker can eavesdrop on the detailed activities and operations that are happening during smartphone charging (e.g., PIN code and text input), even without engaging in data communication. With a trained model, Surf-Snooping exhibits a 100% accuracy on device model identification, while the activity recognition accuracy can reach up to 86.7%, 89.9%, and 81.7% for password recognition, application identification, and keystroke inference, respectively. It is noteworthy that Surf-Snooping achieves accuracies of 92.3%, 98%, and 94.5% for the three types of activity categorization when the phone is fully charged. We also validate the privacy leakage risk of Surf-Snooping with different scenarios, and our work reveals an inherent flaw in the current implementation of wireless charging systems. It provides enhanced obfuscation and stability, requires no physical interference with the charging infrastructure, and remains effective throughout the entire charging cycle.

1. Introduction

In recent years, there has been significant progress in wireless charging technology, with wireless charging stations for smartphones and other devices becoming ubiquitous in various public locations such as hotels, offices, cafes, and airports. Unlike traditional wired charging methods, wireless charging can eliminate the need for connecting cables and reduce the risk of sparks and electric shocks, which provides enhanced safety and convenience. The Qi standard (Van Wageningen and Staring, 2010), established by the Wireless Charging Consortium currently stands as the leading wireless charging protocol for cell phones.

Multi-port USBs commonly serve as interfaces for computers and peripheral hardware devices (e.g., keyboards, mice, USB flash drives), facilitating data communication and transmission. Besides, they are also utilized as power sources for wireless chargers, enabling multiple devices to be powered simultaneously. Nowadays, USB power cords and charging stations are extensively deployed in public spaces, e.g., libraries, cafes, and shopping malls. While existing studies predominantly examine the security of USB interfaces like USB hubs, primarily utilized for data transmission (Tian et al., 2018; Su et al., 2017; Griscioli et al., 2016), neglecting the privacy and security concerns of the multi-port USB as a power source.

In this paper, we introduce Surf-Snooping, a side-channel attack capable of inferring user activities and operations during wireless charging on a cell phone through a nearby USB port, especially when multiple devices are connected simultaneously. As shown in Fig. 1, the vulnerability arises from a circuitry design flaw in USB multi-port power supplies, where multiple ports are connected in parallel, sharing a common 5 V DC supply voltage. During wireless charging, electromagnetic induction causes voltage dynamics in adjacent ports connected in parallel due to user interactions with the device. Monitoring these voltage dynamics from an adjacent port enables the inference of mobile device activities, introducing a risk to the confidentiality of user's private information.

Prior research has demonstrated that the power consumption patterns during the charging process of mobile devices through wired connections may reveal sensitive information (Cronin et al., 2021; Wang et al., 2022). However, fewer studies have delved into the

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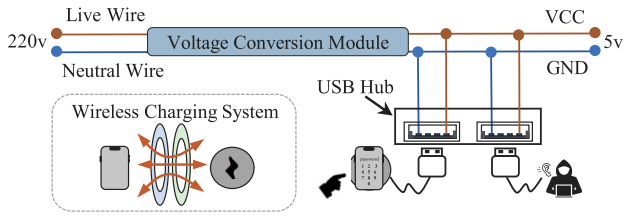


Fig. 1. Illustration of Surf-Snooping.

potential for side-channel attacks during wireless charging. In contrast, Surf-Snooping surpasses existing side-channel attack methods on wireless charging systems (Liu et al., 2022; La Cour et al., 2021; Wu et al., 2021; Ni et al., 2023b) by eliminating the need for hardware modifications or software installations on the charging device or charger. It offers greater stability and feasibility in real-world applications, maintaining high levels of stealth without requiring physical interference with the charging infrastructure. The only thing we need to do is simply plug into a free port. Additionally, Surf-Snooping remains effective throughout the entire charging cycle. To the best of our knowledge, this is the first approach to non-intrusively eavesdrop on private user information via USB power consumption without being dependent on the wireless charging system architecture.

We first validate the feasibility of Surf-Snooping via preliminary experiments and analyze the behind theoretical basis. The Surf-Snooping system consists of three modules: signal preprocessing, feature extraction, and privacy leakage inference. This system discerns user activities on a charging mobile phone by monitoring the dynamic voltage fluctuations of an adjacent port. Specifically, the collected signal undergoes preprocessing to extract features for device identification and targeted privacy inference. Surf-Snooping aims to extract three types of sensitive data from the charging device: unlocking passwords, application launches, and keyboard inputs within the application.

We conduct a comprehensive evaluation of Surf-Snooping using commercially available devices. Our evaluation involves ten popular cell phones, each running on one of the three major operating systems: Android, Harmony, and iOS. Additionally, we utilize five wireless chargers from different brands, along with five multi-port USB power supplies. Our data collection encompasses three categories: numeric password unlocking, application launching, and full-keyboard input. Surf-Snooping achieves impressive accuracy rates: 86.7% for individual numeric password recognition on lock screens, 89.9% for application identification, and 81.7% for recognizing individual alphabets when inferring full keyboard keystrokes within specific applications. We systematically analyze various factors that could impact Surf-Snooping, highlighting its robustness against diverse influences. Our contributions primarily focus on five main areas:

- We introduce a new side-channel attack Surf-Snooping that leverages induced voltage changes at a nearby USB port, resulting from user actions during wireless charging, to infer sensitive information on smartphones.
- We validate the feasibility of Surf-Snooping via preliminary experiments and analyze its theoretical underpinnings.
- We design a three-module system to demonstrate the practicality of Surf-Snooping. Through this system, we successfully infer lock screen passwords, launched applications, and keystrokes.
- We extensively evaluate Surf-Snooping, demonstrating its effectiveness under various impact factors, including remaining phone battery level, phone model, multi-port USB power supply, and wireless charger.
- We discuss Surf-Snooping's distinction between targeted and non-targeted privacy activities in multi-user scenarios, as well as potential defense mitigation measures.

The remainder of the paper is structured as follows: Section 2 introduces the background knowledge of wireless charging and USB, as well as the fundamentals of the Surf-Snooping system. Section 3 demonstrates the feasibility study and threat model. Section 4 provides a detailed overview of the Surf-Snooping system, including specific design and implementation details. In Section 5, we construct an experimental platform to evaluate the effectiveness of the system and present an extensive evaluation of the Surf-Snooping system. Section 6 discusses the implementation in multi-user scenarios, potential mitigations, and a comparison of Surf-Snooping with previous related work, and finally Section 7 concludes and looks forward.

2. Background and principle

2.1. Wireless charging

Currently, there exist four primary methods of wireless charging (Lu et al., 2016): Inductive Charging, Resonant Inductive Charging, Radio Frequency (RF) Charging, and Microwave Charging. Among these methods, electromagnetic induction methods are predominantly utilized in portable devices like cell phones and wearable devices. The wireless charging system, as shown in Fig. 1, consists of two main components (La Cour et al., 2021): the wireless charger and the device being charged (e.g., a cell phone), both equipped with coils. The coil in the charger functions as the transmitter, and the coil in the device acts as the receiver. The transmitting coil, which is connected to a wired power source, converts DC power into AC power using a power inverter unit. This AC power, when passed through the transmitting coil, creates an oscillating magnetic field that induces a voltage in the device's receiving coil. Subsequently, this induced voltage is converted back to DC, either to charge the device's battery or for immediate use by the device, through a rectifier unit. Throughout the charging process, the receiver and transmitter maintain constant communication (Wu et al., 2021), with the receiver sending data packets to the transmitter to regulate the charging power (Van Wageningen and Staring, 2010). The intensity of the charging power is adjusted based on the device's battery level, and varying user activities on the device lead to different rates of power consumption (Yan et al., 2015), necessitating corresponding adjustments in charging power by the transmitter (Cronin et al., 2021).

2.2. The USb interface

USB interfaces are advantageous as they facilitate both data transfer and power supply. This is notably more convenient than employing a range of USB power adapters. USB wall-mounted sockets and charging stations enable the simultaneous charging of multiple devices, a common requirement in residential areas, hotels, airports, and offices. The most prevalent variants of charging ports are USB-A and USB-C. This article focuses on the USB-A port, with the understanding that the discussion applies similarly to the USB-C port.

The conversion of the standard 220 V household AC voltage, as depicted in Fig. 1, into 5 V DC to power a USB-connected charging device through the voltage conversion module involves four critical stages: transformation, rectification, filtering, and voltage regulation. A USB-A charging port incorporates four pins: Pin 1 functions as the ground, establishing a common ground for the circuit; Pins 2 and 3 are designated as data transmission pins; and Pin 4 serves as the power pin. It is common practice to integrate multiple USB power ports within outlets or walls, as shown in Fig. 2, typically connecting them in parallel within the circuit. This configuration ensures that all ports share the same 5 V voltage, making them susceptible to mutual voltage fluctuations, which can lead to signal crosstalk. For instance, voltage variations in a port engaged in wireless charging can induce detectable fluctuations in adjacent ports. An adversary could exploit these variations, specifically the differential in voltage drops across the power (VCC) and ground (GND) lines of neighboring USB ports, to deduce the activities and operations of the charging device, thereby compromising user privacy.

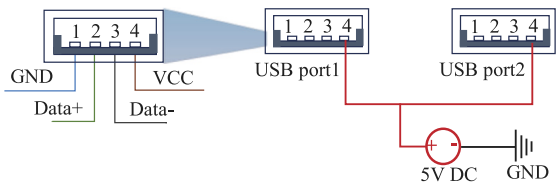


Fig. 2. Structure of a multi-port USB.

2.3. Fundamental principles of Surf-Snooping

When a cell phone is connected to a wireless charger for charging, a two-way interaction of electrical energy occurs. During cellphone operation, power consumption occurs. The energy required is mainly sourced from the battery. However, a portion of the power is directly supplied by the charger. Notably, when the battery is fully charged, all the electrical energy necessary for the cellphone's operation will be directly provided by the charger.

Smartphone activities necessitate the use of specific hardware components. As a device undergoes state transitions or performs different tasks such as unlocking with a password or opening applications, varied workloads incur different power consumptions. (Liu et al., 2022). Consequently, initiating an action on the smartphone causes fluctuations in the load on the receiving coil, leading to current variations. These fluctuations, due to electromagnetic induction between the transmitting and receiving coils, result in corresponding voltage changes in the transmitting coil. Therefore, the voltage dynamics of the wireless charger mirror the smartphone's operations. Design flaw (multiple ports are connected in parallel) in multi-port USB power supplies can reveal these voltage changes at adjacent USB ports. For example, answering a phone call changes the load and thus the voltage. Similarly, different screen animations and levels of CPU usage across applications result in unique power consumption profiles and distinctive voltage patterns.

Since the electromagnetic induction of the transmitting and receiving coils is mutual, wireless charging uses electromagnetic induction to create an electromotive force in the cell phone's receiving coil, which is subsequently transformed into DC voltage to power the device. We denote this DC voltage as $U_C(t)$. The user's operations on the cell phone result in different load changes $\Delta R(t)$, causing a varying current $I(t)$ at the cell phone's receiving coil. The resistance of the receiving coil is R. According to Ohm's law, the varying current in the coil can be expressed as:

$$I(t) = \frac{U_C(t)}{R + \Delta R(t)} \tag{1}$$

The changing current within the coil generates an oscillating magnetic field in space. According to the principle of electromagnetic induction, this field, in turn, induces a corresponding electromotive force (denoted as U_{ind}) in the transmitting coil of the charger.

$$U_{ind} = M\frac{\Delta I(t)}{\Delta t} \tag{2}$$

Here, M represents the mutual inductance coefficient. Thus, the total voltage U_1 measurable by the USB charging port on the charger is the sum of the power supply voltage (denoted as U_P) and the induced electromotive force:

$$U_1 = U_P + U_{ind} \tag{3}$$

Due to different hardware components, the voltage crosstalk signal attenuation rates of different multi-port USBs are different. If the attenuation rate of signal crosstalk between USB interfaces is k, the signal we can measure from the neighboring port is U_2. Substituting into Eqs. (1)(2)(3), we can get:

$$U_2 = k\left(U_P + M\frac{\Delta\left(\frac{1}{\Delta R(t)}U_C(t)\right)}{\Delta t}\right) \tag{4}$$

In our specific scenario, the k value of the same pair of USB ports and the mutual inductance coefficient M are constants. As a 5 V DC power supply, U_P is stable. Therefore, the voltage signals U_2 of adjacent ports that we can measure are closely related to $\Delta R(t)$ caused by mobile phone activities, and the activity privacy of mobile phone devices can also be inferred through the voltage signals.

3. Feasibility and threat model

According to the theoretical framework presented above, we carry out preliminary experiments to assess the feasibility of Surf-Snooping. Additionally, we delineate common scenarios in which such attacks could occur in real-world situations.

3.1. Feasibility study

In this section, we illustrate an example of launching Surf-Snooping on a smartphone using a multi-port USB hub. Charge a HUAWEI Mate40 smartphone using a wireless charging pad that connects to a USB hub via a cable. This USB hub contains multiple USB ports with power supply capability. We maintain The phone's battery level between 40% and 60%. The user executes three operations: unlocking the phone with passwords, opening WeChat, and utilizing WeChat's full keyboard to send messages. Additionally, we monitored the voltage changes of two ports, the charging port and its adjacent port with an oscilloscope.

Fig. 3 presents the voltage signals from two USB ports, reflecting different user activities. We use a moving average filter to enhance the visual distinguishability of the signal. Initially, digits 1, 2, 3, and 4 were sequentially entered on the mobile phone's unlock screen, with a one-second pause between each entry. Following this, within the chat interface of WeChat, keys 'a', 'b', and 'c' were pressed on the 26-key keyboard, each button press resulting in a unique voltage fluctuation pattern, as indicated by the signal's distinctive peaks and troughs. The figure also reveals how signal crosstalk between the two USB ports causes concurrent yet divergent changes in the signals, with the charging port represented by a gray curve and the adjacent port by a blue curve. Moreover, when the phone is idle, the voltage across the receiving coil remains stable. When the same model of the device performs three activities, the signal traces undergo highly similar dynamic changes. However, the same operation produces different changes on different models. Our feasibility study demonstrates the presence of signal crosstalk leakage between USB ports, further demonstrating the possibility of privacy inference.

3.2. Threat model

Wireless charging stations, equipped with USB ports, have become increasingly common in public spaces like cafes, hotels, and airports. As shown on the left side of Fig. 3, this study investigates a scenario in which an individual charges their smartphones using either a public wireless charger or a personal charger connected to a public USB power outlet, often found with multiple ports in walls or on charging stations. During charging, users engage in various activities on their phones, such as unlocking them or sending messages. An attacker, using a USB data cable, pretends to charge at an adjacent USB port and covertly records the voltage signal from the port. The dynamic voltage signal is analyzed to deduce the user's activities on their mobile device, compromising their privacy and making detection challenging for victims. Furthermore, due to the absence of security authentication in the current wireless charging protocol, an attacker can secretly record the voltage trace from the USB interface and analyze the user's private data either online or offline.

We take the airport scenario as an example. In an airport's shared power outlet setup, an attacker can monitor the voltage signals of adjacent ports by connecting a device to an adjacent port and posing

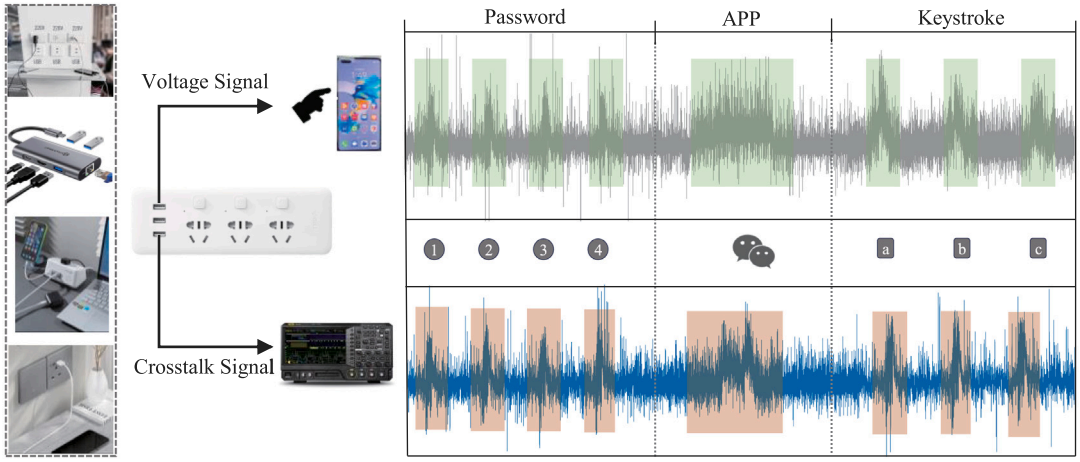


Fig. 3. Crosstalk leakage of voltage dynamics induced by password inputting, application launching, keystroke pressing from two adjacent ports.

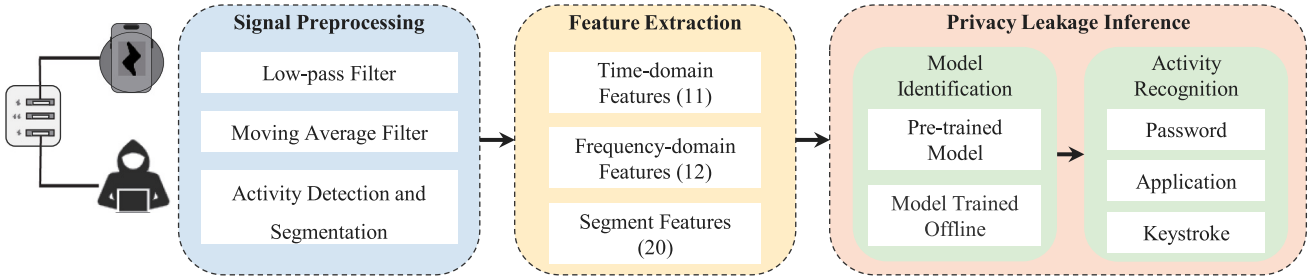


Fig. 4. The workflow of Surf-Snooping. Surf-Snooping includes three modules: (1) signal preprocessing, (2) feature extraction, (3) privacy leakage.

as a regular charging user. Attackers may deploy sophisticated "bait" chargers. Externally, these chargers are indistinguishable from normal ones, yet they are internally modified maliciously. They conceal the attacking device within normal charging devices and, using a wireless communication module, transmit the collected voltage data to a remote receiving device for analysis and processing.

The attacker does not require knowledge of the victim's mobile phone model or any related information. Nor is there a need to inject attack instructions into the mobile phone or manipulate the charging pile. Attackers can train classification models for mainstream cell phone models in advance. With prior knowledge of common smartphone voltage signal patterns and a trained classification model, the attacker can conduct the attack covertly. This process does not mandate the victim's authorization, and voltage detection does not impact the phone's power consumption during charging. As a result, the attack is non-intrusive and stealthy.

4. Surf-Snooping design

In this section, we present an overview of the Surf-Snooping system, and we also discuss its design and implementation details respectively.

4.1. System overview

Fig. 4 illustrates the workflow of Surf-Snooping. During signal acquisition, use integrated or off-the-shelf signal acquisition devices (e.g., oscilloscopes) to acquire adjacent port voltage signals at a moderate sampling rate. Avoid signal distortion caused by too low a sampling rate and excessive data size caused by too high a sampling frequency. As described in Section 5, we set the sampling frequency of the oscilloscope to 2 MHz in our experiments. The collected data undergoes signal preprocessing, feature extraction, and privacy leakage inference. The first module comprises a low-pass filter, a moving average filter, and activity detection and segmentation. Subsequently, feature extraction is conducted on the preprocessed signal segments, extracting time domain, frequency domain features, and segmentation features for classification training. We can therefore infer that leaked privacy encompasses unlocking passwords, launched applications, and keystroke inference within applications with the privacy leakage inference module.

4.2. Signal preprocessing

The signal preprocessing module filters noise from adjacent port voltage signals and performs detection and segmentation of active signals. It prepares activity signal samples with trigger events of specific lengths for the subsequent feature extraction module.

We use software filters to process the signals. Variations in USB power components and supply voltages can introduce noise interference in the collected voltage signals, potentially overwhelming the active signal. To mitigate this, we utilize filters to eliminate unwanted frequency interference. We use the signal processing component of Matlab to build filters. Specifically, a first-order low-pass filter with a cutoff frequency of 0.1 MHz is implemented to remove DC offset from the original signal. Despite these filtering efforts, the signal still exhibits multiple high-amplitude peaks. To tackle this issue, we implement a moving average filter. This entails applying a sliding window of N sample points. We calculate the average value of consecutive data points within it, which is then assigned as the value of the new data point, reducing signal jitter. This process can be formulated as:

$$y_i = \frac{1}{N}\sum_{k=0}^{N-1} x_{i+k} \tag{5}$$

where y_i is the mean of the jth window and x_{i+k} denotes the value of the kth sample after the current sample point x_i in the window. $Y = (y_1, y_2, y_3...)$ makes up the new sample signal.

To detect and segment the active part containing the triggering event, we utilize a sliding window approach to partition the signal sequence Y. The sliding window size is set to ω with a step size of $\omega/2$ and a threshold τ is established, where segments exceeding this threshold are considered active. For each window i, compute the variance within the window:

$$\mathrm{Var}_i = \frac{1}{\omega} \sum_{j=i}^{i+\omega-1} (y_j - \bar{y}_i)^2 \tag{6}$$

$\bar{y}_i$ is the mean value for window i, when$\mathrm{Var}_i > \tau$, it is marked as the start of the activity.

4.3. Feature extraction

Following signal preprocessing, multiple samples encapsulating active events are obtained. For each sample, a predetermined set of features is extracted to construct a feature vector. These features include statistical measures of time-domain signals and frequency-domain features obtained through Fourier transform (FFT). For the former, we compute the mean, standard deviation, time integral, root mean square (RMS), peak-to-peak value, skewness, kurtosis, crest factor, clearance factor, shape factor, and pulse index for each sample. As for the latter, besides the 12 statistical characteristics of the frequency-domain signal, we partition the frequency range into 20 equally sized segments and compute the average amplitude of all frequencies within each segment, using it as a feature. Consequently, we obtain a feature vector comprising 43 elements. Z-score standardization is then applied to the feature vector to mitigate the impact of excessively large features on subsequent classification:

$$\mu = \frac{1}{n} \sum_{i=1}^{n} f_i \tag{7}$$

where f_i is the value of the ith element, μ is the mean value, and n is the number of elements of the feature vector respectively. The Z-score z_i for each element f_i can be expressed as:

$$z_i = \frac{f_i - \mu}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(f_i - \mu)^2}}, i \in [1, n] \tag{8}$$

The standardized feature vector $Z = (z_1, z_2, \dots, z_n)$ is utilized for subsequent privacy content inference.

4.4. Privacy leakage inference

After extracting the necessary features, in the privacy leakage inference module, we first infer the cell phone device model used by the victim. For a specific phone model, three types of activity recognition are performed: password, application, and keystroke.

4.4.1. Phone model identification

To infer sensitive information during the charging process, it is essential to identify the victim's cell phone model. This can be achieved by pre-training classification models for popular cell phone models in the market, or by recording signals in actual attack scenarios for subsequent offline inference training.

When the phone is idle, the voltage across the receiving coil remains stable. Various internal components such as CPU, GPU, and screen drivers, along with screen technology and resolution, contribute to distinct power consumption patterns. Even for devices with minimal hardware disparities, such as various iPhone models sharing similar architectures, subtle hardware differences persist. These devices employ different chips and feature distinct screen sizes. For instance, the iPhone 14 is equipped with an A16 Bionic chip, whereas the iPhone 13 utilizes an A15 chip. Moreover, the iPhone 14 and iPhone 14 Plus have screen sizes of 6.1 inches and 6.7 inches, respectively. Henceforth, various cell phone models exhibit diverse frequencies and amplitudes in their underlying voltage signals. Therefore, we collect T_{temp} second voltage signals from mainstream cell phone models as templates, and each signal contains $T_{temp} \times F$ sample points, where F is the sampling rate. Thus, the large amount of data collection inherent in traditional machine learning methods is avoided. Due to variations in power levels, a single cell phone model may display different base voltages. Therefore, we choose to implement baseline correction by subtracting the minimum value $\min(V)$ from each signal $V = (v_1, v_2, \dots, v_{T_{temp}\times F})$. This process can be formulated as follows:

$$c_i = v_i - \min(V), i \in \left[1, T_{temp} * F\right] \tag{9}$$

The signal after removing the baseline drift is $C = (c_1, c_2, \dots, c_{T_{temp}\times F})$.

We segment the signal C into 50 smaller samples, utilizing the 23 time-domain and frequency-domain features described in Section 4.3 as feature vectors and applying the Random Forest algorithm (RF).

We collect voltage stability signals from prevalent mobile phone models and pre-train the model to identify the victim's device model, facilitating subsequent privacy inference. Since the victim's cell phone model may not belong to the pre-trained models collected beforehand, the classification model of device recognition further checks the confidence value of each output class when inferring the cell phone model. If the confidence levels are all low, indicating that the model is not available in the database, we can record the voltage signals for offline training and analysis. The newly recognized device models are then incorporated into our database.

4.4.2. Activity recognition

After identifying the device model, we proceed to recognize the three activities performed by the user on the device. Our activity recognition approach is systematically divided into two main phases: training and testing. During the training phase, Principal Component Analysis (PCA) reduces the dimensionality of feature vectors extracted from signal samples. To infer sensitive information such as passwords and applications, we utilize a Random Forest classifier. Moreover, to improve the accuracy of full-keyboard keystroke inference, we adopt an integrated classifier approach incorporating a voting mechanism. This mechanism combines multiple classifiers, including Random Forest, Support Vector Machine, K-Nearest Neighbor, and Linear Discriminant. The final classification result was determined through a voting mechanism. We use the GridSearchCV tool in Scikit-learn for parameter search and model selection. Finally, we evaluate the model performance using ten-fold cross-validation.

5. Evaluation

This section outlines the development of our experimental platform, and the process of data collection, and assesses the efficacy of Surf-Snooping.

5.1. Experimental methods

Experimental Setup: As shown in Fig. 5, to evaluate the overall performance of Surf-Snooping, a Ugreen USB hub with multiple USB charging ports is used. A Xiaomi wireless charger is connected to one port to charge the Huawei Mate40 smartphone, while the voltage at the other port is recorded using a RIGOL MSO5000 digital oscilloscope to collect data samples. We perform the following operations: (i) unlocking the smartphone by entering a password; (ii) launching different mobile applications; and (iii) typing text in a chat application such as WhatsApp. A cut-off USB cable connects to the other port. To detect crosstalk leakage of voltage signals on the power line, the oscilloscope's passive probe measures the voltage on the VCC line relative to the GND line on the cut-off USB cable. We divide the experimental process

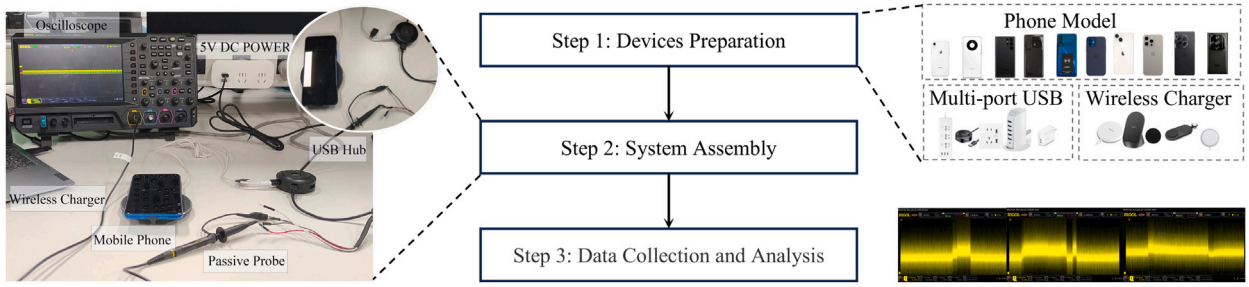


Fig. 5. Experimental Setup.

into three steps. In the device preparation phase, we select multiple wireless charging devices, chargers, and USB power supplies. During the system assembly phase, each wireless charging device is combined with a corresponding charger and power supply to form a wireless charging system. The charger is connected to the power supply to deliver power to the wireless charging device. In the data collection and analysis phase, we set the oscilloscope's sampling frequency to 2 MHz and connect a passive probe to capture crosstalk leakage signals. Signal samples from each system are collected and stored in the dataset. We follow the same procedure to experiment with five battery levels for charging devices (Section 5.3.1), ten mobile devices from different vendors (Section 5.3.2), five multi-port USB chargers (Section 5.3.3), and five different brands of wireless chargers (Section 5.3.4).

As depicted in Fig. 5, our experimental setup includes ten distinct models of mainstream cell phones: iPhone XR, iPhone12, iPhone14, iPhone15, OnePlus12, OPPO Find X7Pro, HUAWEI Mate40, Samsung Galaxy S22 Ultra, Xiaomi13, and HUAWEI nova6 (requiring a wireless charging adapter as it lacks built-in wireless charging capability). Additionally, we utilize five multi-port USB power supplies sourced from various vendors, namely Bull, Ugreen, Legrand, Tessan, and Belkin, each with different architectures. Furthermore, we incorporate five different brands of wireless chargers, including Samsung, HUAWEI, Xiaomi, Belkin, and iPhone. Specific information about the hardware devices in the experiment is shown in Tables 1 and 2.

Data Collection[1]**:** To control for other variables, study the performance of Surf-Snooping between different models of devices. Throughout the data collection process, all phones maintain their basic settings, with battery levels consistently kept between 40% and 60%, except as noted in Section 5.3.1, where we will discuss the impact of battery power on performance. Stable signals are gathered from ten mobile phone models in the locked screen state to identify their models. For password unlocking, ten-digit numbers ranging from 0 to 9 are collected. Application identification involves launching each of the ten currently popular apps. Keystroke inference includes pressing each of the 26 letters from 'a' to 'z'.

5.2. Overall effectiveness

We assemble a prototype system using a HUAWEI Mate40, a Ugreen USB hub, and a Xiaomi wireless charger. Throughout the process, the phone battery is maintained at 40% to 60%. We evaluate the overall performance of Surf-Snooping in identifying phone models and conducting three privacy inference activities.

Phone model identification: We initiate our analysis by evaluating the precision of device identification. Utilizing a Random Forest Classifier, we train the model and determine the importance of features. As shown in Fig. 6, the identification accuracy for all ten mobile phone device models achieves 100%. Following this procedure, we evaluate the performance of three distinct activities.

[1] https://github.com/houyue355/Experimental-Data

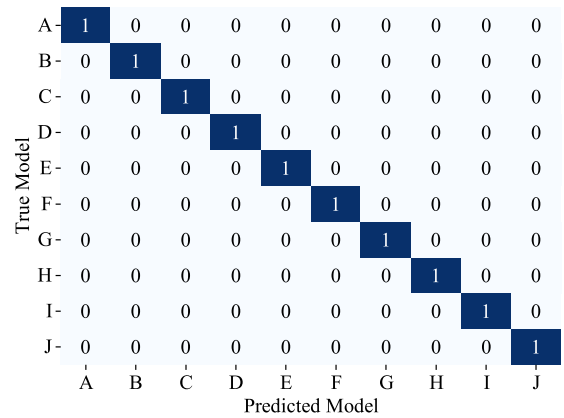


Fig. 6. Confusion matrix of ten different cell phone model classifications. A: iPhone XR, B: HUAWEI Mate40, C: Samsung Galaxy S22 Ultra, D: Xiaomi13, E: HUAWEI nova6 with a wireless adapter, F: iPhone12, G: iPhone14, H: iPhone15, I: OnePlus12, J: OPPO Find X7Pro.

Passwords: we collect 100 samples for each ten-digit number from 0 to 9, with a 1-second interval between each press. We use ten-fold cross-validation to evaluate the performance of the password recognition model. The evaluation results' confusion matrix is depicted in Fig. 7(a). The overall accuracy of unlocking password recognition stands at 86.7%. Notably, most misclassifications arise between two adjacent keys. For instance, on the numeric soft keyboard, misclassifications between '1' and '2'/'4' account for 10%.

We choose 500 random 4-digit combinations to test the classifier. For each digit in a four-digit password, the top two classifications of this confidence value are formed into a vector at the time of classification. For a four-digit password, this corresponds to four vectors, each containing two elements of the top two of the respective confidence values. Afterward, the password sequence is constructed using the combination. The results show that 92.4% of the passwords were inferred accurately within ten combination experiments. Our cell phone systems usually allow us to enter the correct password within ten times or less, which is a big threat in practice.

Applications: We launch each of the ten currently popular apps (Alipay, Amazon, Facebook, Google Maps, Amap, Tencent QQ, MicroBlog, WeChat, TikTok, and NetEase) 50 times. After the application is loaded, we wait for two seconds for the signal to stabilize, clear the background running programs, and proceed to the next launch. Similarly, we evaluate application inference using ten-fold cross-validation. The resulting confusion matrix in Fig. 7(b) attains a classification accuracy of 89.9% for the ten applications. As we can see from the figure, there is always a misclassification between Google Maps and AMap. This may occur because both need to read the location information in the background when launching the application to display the map. Also, similar misclassifications were seen between Alipay and Facebook, which may be related to the black-and-white pixels they display,

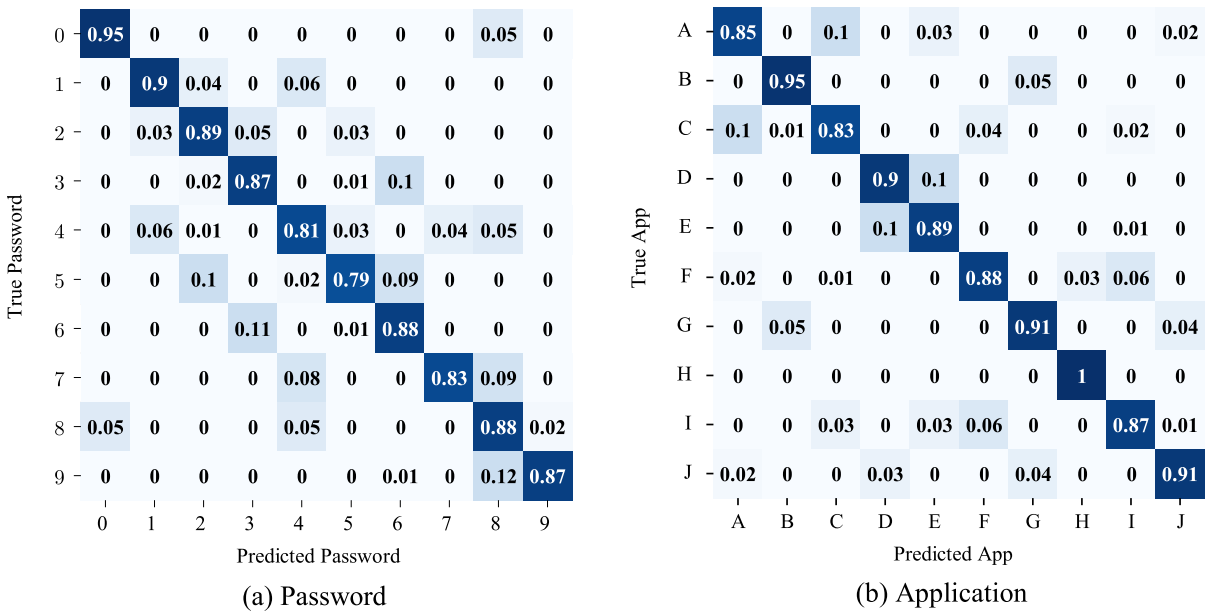


Fig. 7. Overall effectiveness of Surf-Snooping on HUAWEI Mate40. (a): Confusion matrix of password recognition (from 0 to 9). (b): Confusion matrix of 10 apps identification. A: Alipay, B: Amazon, C: Facebook, D: Google Maps, E: Amap, F: QQ, G: MicroBlog, H: WeChat, I: TikTok, J: NetEase.

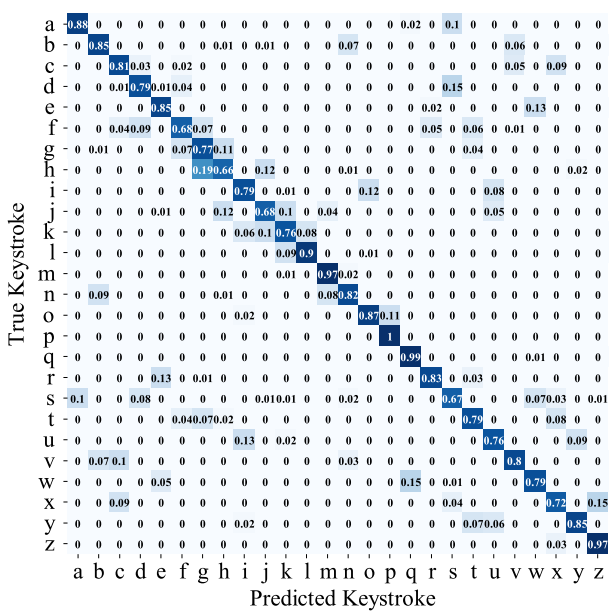


Fig. 8. Confusion matrix of keystroke inference (from "a" to "z") on HUAWEI Mate40.

and CPU consumption. Previous studies have shown that the display (i.e., touchscreen) and CPU/GPU significantly impact the overall power consumption of a smartphone (Carroll and Heiser, 2010). Screen refresh cycles and resolutions vary among devices. These differences result in distinct power-consumption patterns when the same video is played in a device's window. Specifically, when the mobile operating system (OS) captures a user input action, it provides visual feedback by rendering and drawing an animation on the screen. This process causes pixels to change colors rapidly and induces significantly different voltage states among different devices.

Keystrokes: For keystroke inference, we press each of the 26 letters 'a' to 'z' 50 times solely for training purposes, with a second interval between each key press. Due to individual differences in key-pressing habits, we select 30 words and enlist 5 volunteers to input them through the keyboard as test data samples. Additionally, we add some individual keystroke samples to balance the test samples. Approximately 1240 samples collected from five volunteers serve as test data. Fig. 8 depicts the confusion matrix of key inference, achieving an overall accuracy of 81.7%. Moreover, Surf-Snooping exhibits higher accuracy in identifying the edge keys of the full keyboard, such as 'q' (99%), 'p' (100%), and 'm' (97%). Similar to the numeric soft keyboard, incorrect key recognition often occurs between two adjacent keys.

Research indicates that the display (i.e., touch screen) and CPU/GPU significantly contribute to the overall power consumption of smartphones (Carroll and Heiser, 2010). When buttons on the touch screen are pressed, adjacent ones typically generate comparable black and white pixels (Cronin et al., 2021), resulting in similar energy consumption and load changes as expressed in formula (4). Consequently, our signal samples exhibit similar voltage trajectories.

By exploiting Surf-Snooping, an adversary can accurately discern the victim's unlocking password and subsequently gain unauthorized access to the victim's smartphone while it is being charged, thereby compromising additional user privacy. Moreover, through application and keystroke inference, the attacker can further scrutinize the victim's lifestyle and activities. Particularly concerning are payment applications, where the attacker may obtain the victim's payment password, address, and other sensitive information.

5.3. Discussion

To ascertain the practicality of Surf-Snooping across diverse environments and under various influencing factors, we performed a series of experiments aimed at assessing how classifier accuracy is affected by the following variables: (1) remaining phone battery level, (2) phone model, (3) Multi-port USB power supply, (4) wireless charger, as shown in Tables 1 and 2, following identical steps and strategies as before for data collection and evaluation across three activities: password recognition, application identification, and keystroke inference.

5.3.1. Impact of different remaining battery levels

The charging device of the victim may have varying remaining battery levels in different usage scenarios. Therefore, we categorized battery power into five levels (0%–20%, 20%–40%, 40%–60%, 60%–80%, and 80%–100%) to assess its influence on classification accuracy. We use the experimental setup described in Section 5.1 (a Ugreen USB

Table 1
The overall performance of Surf-Snooping of different phone models.

Brand	System	Password	App	Key stroke
iPhoneXR	iOS	74.2%	76.6%	71.9%
iPhone12	iOS	68%	72.3%	66.9%
iPhone14	iOS	74.6%	75.1%	71.1%
iPhone15	iOS	78.4%	79.2%	73.8%
HUAWEI Mate40	Harmony	86.7%	89.9%	81.7%
HUAWEI nova6[a]	Harmony	86.8%	92.1%	85.4%
Xiaomi13	Android	85.3%	88.2%	83.7%
OnePlus12	Android	79.8%	81.4%	76.5%
Samsung Galaxy S22 Ultra	Android	78.6%	82.4%	75.9%
OPPO Find X7Pro	Android	80.5%	84.7%	78.4%
Average Accuracy		79.3%	82.2%	76.5%

[a] HUAWEI nova6 requires a wireless charging adapter as it lacks built-in wireless charging capability.

Table 2
Hardware devices used for Surf-Snooping performance evaluation.

Multi-port USB			Wireless Charger	
Brand	Ports Number	Type	Brand	Power
Bull	3	Power Strip	Samsung	30 W
Ugreen	4	USB Hub	HUAWEI	80 W
Legrand	2	Panel Socket	Xiaomi	50 W
Tessan	5	USB Hub Base	Belkin	15 W
Belkin	2	Charger	iPhone	15 W

hub, a Xiaomi wireless charger, and a Huawei Mate40 smartphone), and varying remaining battery levels of the cellular device, to evaluate the effect of Surf-Snooping at different battery levels. As illustrated in Fig. 9(a), with increasing battery levels, the accuracy of identifying lock screen passwords, applications, and keyboard keystrokes also increases. When the battery level surpasses 80%, the accuracy of privacy leakage reaches its peak, with password recognition accuracy at 89.1%, application classification accuracy at 91.2%, and keyboard keystroke inference accuracy at 83.2%. Experimental results indicate that higher battery levels pose a greater risk of privacy leakage. However, even at lower battery levels, Surf-Snooping can achieve the desired privacy breach.

It is noteworthy that Surf-Snooping achieves accuracies of 92.3%, 98%, and 94.5% for the three activity classifications when the phone is fully charged. However, it is common for cell phone users to keep their devices charged even when fully charged. In such instances, the power supply directly fuels the user's operations, bypassing the need to recharge the battery. Consequently, any dynamic changes in the voltage signal directly mirror the user's activities.

5.3.2. Impact of different phone models

Different smartphones employ varying energy consumption and management strategies at both hardware and software levels. Even within the same brand, different models of mobile phones exhibit significant differences. To assess the impact of different mobile phone models on Surf-Snooping, we conducted evaluations on ten diverse charging device models, including iPhone XR, iPhone12, iPhone14, iPhone15, OnePlus12, OPPO Find X7Pro, HUAWEI Mate40, Samsung Galaxy S22 Ultra, Xiaomi13, and HUAWEI nova6 with a Wireless Charging Adapter (Qi-Wireless-Charging.net, 2020). We use the experimental setup described in Section 5.1 (a Ugreen USB hub, a Xiaomi wireless charger), and varying phone models, to evaluate the effect of Surf-Snooping across different phone models. The battery level of each type of cell phone is maintained between 40% and 60%. Table 1 illustrates that the classification model for these charging devices achieves an average accuracy of 79.3% in lock screen password recognition, 82.2% in application classification, and 76.5% in full keyboard key inference. Different phone models exhibit different performance. This discrepancy may arise from variations in screen technologies and keyboard layouts among the devices.

5.3.3. Impact of multi-port USb power supplies

Due to the diverse internal circuits found in various multi-USB port power supplies, the resulting crosstalk voltage signals exhibit varying patterns. To assess the applicability of Surf-Snooping across a range of multi-USB port power supplies, we additionally collect data from five different brands (Bull, Ugreen, Legrand, Tessan, and Belkin) having different structures (wall-embedded, plug-in charging pile, vertical charging pile). We use the experimental setup described in Section 5.1 (a Xiaomi wireless charger, a Huawei Mate40 smartphone), and varying multi-port USB power supplies, to evaluate the effect of Surf-Snooping across different multi-port USB power supplies. The battery level of the cell phone is maintained between 40% and 60%. Fig. 9(b) demonstrates high accuracy across five different multi-USB port power supplies, with lock screen password accuracy ranging from 84.2% to 88.6%, application classification accuracy from 86.4% to 92.1%, and full keyboard key inference accuracy from 80.9% to 85.4%. Moreover, our experiments reveal that the Tessan multi-bit USB power adapter yields lower eavesdropping accuracy. Unlike other USB power supplies, the Tessan has five USB ports, more than any other power supply. This is attributed to the adapter's inherent instability in providing a basic voltage, resulting in increased noise and significant fluctuations in the signal, which obscures the voltage changes induced by user activities.

5.3.4. Impact of wireless chargers

We test Surf-Snooping with five wireless chargers from different brands (Samsung, HUAWEI, Xiaomi, Belkin, and iPhone) and compare the results with the Xiaomi charger used previously. We use the experimental setup described in Section 5.1 (a Huawei Mate40 smartphone, a Ugreen USB hub), and varying wireless chargers, to evaluate the effect of Surf-Snooping across different wireless chargers. The battery level of the cell phone is maintained between 40% and 60%. As depicted in Fig. 9(c), Surf-Snooping demonstrates high accuracy across all five different wireless chargers. The accuracy of lock screen password inference ranges from 76.5% to 86.7%, application classification ranges from 79.4% to 89.9%, and full keyboard key inference ranges from 72.1% to 84.2%. Notably, the iPhone charger exhibits the lowest accuracy among the three privacy inference tasks and displays higher security levels.

5.3.5. Impact of other factors

Environmental factors such as phone brightness and the angle of the charging pad primarily affect the baseline voltage of crosstalk voltage signals rather than their relative dynamic fluctuations. Despite this, Surf-Snooping remains effective in detecting voltage variations caused by user actions. This demonstrates that while these environmental factors impact the absolute value of the voltage signal, Surf-Snooping can adapt to these variations, maintaining its ability to monitor and identify user activities. In summary, Surf-Snooping shows a high level of resilience against various influences.

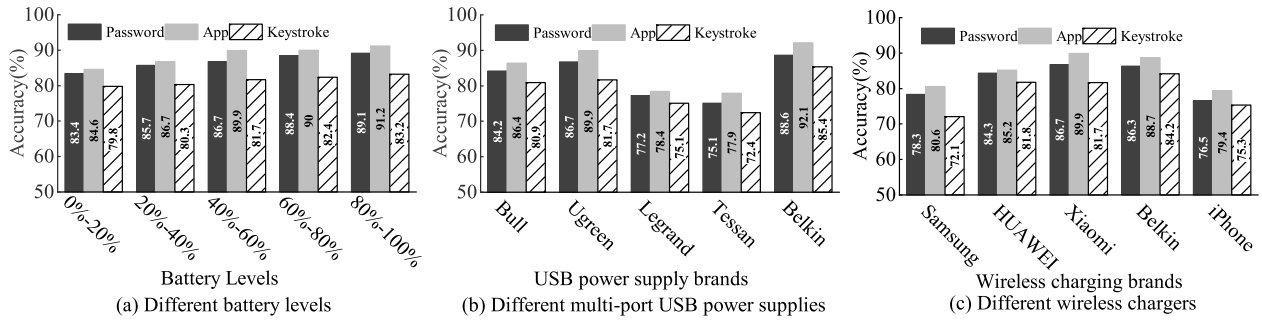
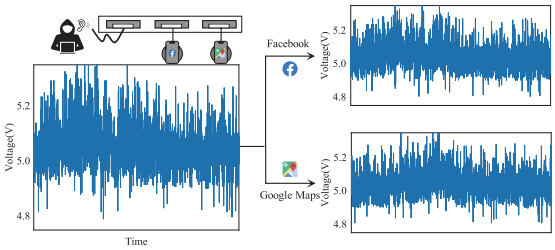


Fig. 9. The overall performance of Surf-Snooping of four practical impact factors: (a) Impact of different battery levels of the phone, (b) Impact of different multi-port USB power supplies, (c) Impact of different wireless chargers.

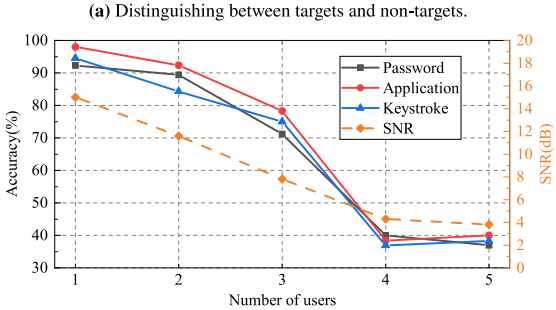
Table 3
Comparison with existing wireless charging work.

Work	Target device	Stability	Valid at any time	Non-intrusive	Attack		
					PIN	APP/Web	Keystroke
Wu et al. (2021)	A hidden adversary coil	Medium	○	○	○	●	○
La Cour et al. (2021)	Power line	High	○	○	○	●	○
EM-Surfing (Liu et al., 2022)	Power line	High	●	○	●	●	●
WISERS (Ni et al., 2023b)	Acoustic and magnetic sensors	Medium	●	●	●	●	●
Surf-Snooping	Multi-port USB power	High	●	●	●	●	●

Legend: ●: 'Yes', ○: 'No'.



(a) Distinguishing between targets and non-targets.



(b) Effectiveness of eavesdropping on multiple users.

Fig. 10. Evaluation of attacking multiple users.

6. Discussion

6.1. Attacks on multiple users

In real-world scenarios, multiple users may simultaneously use a charging station, necessitating the differentiation between targets and non-targets. To address this, we use a Ugreen USB hub to connect two wireless chargers(Xiaomi) to ports 1 and 2, powering the iPhone 12 and HUAWEI Mate40, respectively. During the experiment, the phone battery is maintained at 100%. The voltage signals from the remaining USB port (port 3) are monitored. Both phones simultaneously run different applications, namely Facebook and Google Maps. Given that voltage leakage is a one-dimensional signal, we apply blind source separation (Ni et al., 2023a) to isolate the mixed voltage signals into distinct signals corresponding to each user's activity. Fig. 10(a) demonstrates the separation of the mixed voltage leakage (SNR=11.6 dB) into individual signals when Facebook (SNR=10.2 dB) and Google Maps (SNR=10.4 dB) are launched on the two charging smartphones. We investigate the impact of pixel-related operations on Surf-Snooping. We play a video on an iPhone 12 and launch an application on a HUAWEI Mate 40. After separating the signals, we find that the signal-to-noise ratio of the app-launch signal is high, and the original signal is nearly restored.

To find out at what point Surf-Snooping breaks down, we increase the number of user devices being charged from 1 to 5 to examine the effect of noise in multi-user environments. We simulate a real-life situation where we charge five devices simultaneously and use them. As shown in Fig. 10(b), with the increase in the number of users, the mutual interference among the signals of USB ports becomes more severe. As shown by the orange dotted line, the average signal-to-noise ratio (SNR) of the signals decreases from 15 dB to 3.8 dB. Correspondingly, the accuracy of the three types of privacy inference also drops. Notably, when the number of user devices exceeds 3, the accuracy of Surf-Snooping declines rapidly. We will defer the improvement of Surf-Snooping's performance and the investigation of other potential defense measures in complex multi-user device scenarios to future research.

6.2. Mitigation strategies

Our study demonstrates that an attacker can infer a user's private data by analyzing the voltage dynamics of a wireless charger. To

mitigate the security risks associated with dynamic voltage leakage, various countermeasures can be implemented, broadly categorized into hardware-based and software-based approaches. On the software side, since Surf-Snooping exploits captured voltage signals, an effective strategy is to introduce random noise into the system. This can be achieved by generating dummy traffic packets (Li et al., 2022; Ni et al., 2023a), which are extraneous signals unrelated to actual user activity. Dummy traffic packets are digital constructs. When they are generated and sent within the device's communication stack, they cause changes in the power consumption of the device. As a result, these digital packets indirectly affect the analog voltage signals on the VCC line. These packets obscure voltage patterns, making it more challenging for an attacker to extract meaningful information without affecting the user experience. Additionally, users can reduce privacy risks by installing software on their phones that incorporate these protective features.

On the hardware side, an effective method to mitigate dynamic voltage leakage is to attenuate the high-frequency components that often cause this issue. Wireless charger manufacturers can achieve this by incorporating a simple RC (resistor-capacitor) circuit into the charger. The resistor and capacitor work together to smooth out rapid fluctuations in the voltage signal, thereby reducing the amount of extractable information. Additionally, since the leakage of keystroke information depends on keypad location, handset manufacturers can further mitigate this risk by randomizing the keypad layout. Combined, these hardware and software strategies offer a robust defense against the risks posed by dynamic voltage leakage, ensuring that user data remains secure even against sophisticated eavesdropping techniques.

Protection strategies can be adopted from the charger and client sides. On the charger side, charger manufacturers can optimize the hardware-based circuit design. They can separate sensitive and power circuits and enhance electromagnetic shielding through the use of a multi-layer shell to minimize signal radiation. In the software aspect, they are required to develop intelligent power management algorithms. These algorithms should monitor parameters such as voltage, current, and others, and adjust the charging strategy upon detecting anomalies. On the client side, operating system developers should incorporate charging monitoring and protection features. The operating system is then tasked with monitoring voltage and current levels, alerting users when anomalies are detected, providing relevant suggestions, and managing power consumption by restricting high-power non-essential background programs. Moreover, application developers should optimize graphic rendering algorithms and utilize data caching technology during the application design phase. This approach reduces the impact of applications on charging voltage signals, alleviates the loads on the GPU and CPU, and stabilizes the device's power consumption during charging.

To further mitigate the security risks associated with dynamic voltage leakage, the use of Trusted Execution Environments (TEEs) (Muñoz A. Rios et al.) and TPM technology (Gurgens et al., 2007) should be considered. Integrating TPM modules into wireless charging devices can securely store sensitive data, ensuring protection even if the device is compromised. Additionally, employing S&D design patterns during the system design phase can enhance the system's resilience to a broad range of known security threats and improve its ability to withstand unforeseen attacks.

6.3. Application to other wireless devices

Although this paper concentrates on the privacy issues surrounding smartphone wireless charging, the attack principle of Surf-Snooping is equally applicable to other wireless-charging-enabled devices, including wearables and Internet of Things (IoT) systems. Wearables, like smartwatches and fitness trackers, as well as IoT devices such as smart home appliances and sensors, are increasingly equipped with wireless charging functions and connected to multi-port USBs. By applying Surf-Snooping to these devices, attackers can monitor voltage fluctuations at adjacent USB ports. This enables them to infer user activities and acquire private data, such as movement trajectories and health information.

6.4. Comparison with existing work

We compare Surf-Snooping with four current wireless charging side-channel attacks. As shown in Table 3, Wu et al. (2021) exploit a vulnerability in the Qi standard by intercepting Qi messages using a concealed adversarial coil, yet this method lacks covert operations due to the coil's visibility. La Cour et al. (2021) deduced private information, such as visited web pages, by monitoring current fluctuations in the charging cable. However, this technique is limited to scenarios where the battery level exceeds 80%. EM-Surfing (Liu et al., 2022) utilizes the principle of electromagnetic induction in wireless charging to monitor the voltage signals at the charging port. Compared to ours, (Liu et al., 2022) and (La Cour et al., 2021) both require physical tampering with the charging apparatus to embed a malicious device (e.g., an external resistor on the power line), WISERS (Ni et al., 2023b) performs fine-grained privacy inferences by capturing physical phenomena, electromagnetism and sound generated during the charging process through the phone's built-in sound and magnetic sensors. However, the physical phenomena in Wu et al. (2021), Ni et al. (2023b) can easily be susceptible to interference by the surrounding environment or by human beings, and thus have a lower stability. Our Surf-Snooping has higher stealth and stability, does not require physical interference with charging infrastructure, and is effective throughout the charging cycle. To the best of our knowledge, our work is the first to disengage from the wireless charging system architecture to non-intrusively eavesdrop on private user information via USB power consumption.

6.5. Other side-channel attacks

Cell phones, being the most commonly used electronic devices in daily life, store a vast amount of personal and private information. Considerable attention has been given to side-channel attacks on cell phones (Neugschwandtner et al., 2016; Ba et al., 2020; Owusu et al., 2012; Zhou et al., 2018). While eavesdropping on a user's privacy through hardware components like sensors or speakers within a cell phone has been extensively studied. However, it is not very feasible in reality as it requires physical access to the victim's phone and training the model. Various user operations on the phone result in distinct power consumption patterns, which can be analyzed to infer sensitive information. From inside the cell phone, mobile applications have access to the battery level or usage, and user operations can be inferred by analyzing event logs or power profiles recorded by these applications (Ma et al., 2013; Brouwers et al., 2014; Chen et al., 2017; Michalevsky et al., 2015). From outside of the cell phone, side-channel attacks with power traces can eavesdrop on user privacy through USB power cables, USB ports, or wireless chargers (Cronin et al., 2021; Spolaor et al., 2017; Yang et al., 2016). For instance, as demonstrated by Cronin et al. (Cronin et al., 2021), keystrokes in different screen areas consume varying energy amounts, they measured voltage across a resistor connected to the power cable to infer passwords entered by multiple users. Unlike these methods, Surf-Snooping does not require any hardware physical modification or software installation.

USB, as the most commonly used interface for connecting peripheral devices, provides both data transfer and power supply functions. Prior research has primarily focused on attacking USB data transmission (Neugschwandtner et al., 2016; Su et al., 2017; Monaco, 2018), and even if defenses are developed (Tian et al., 2016; Tian et al., 2015), eavesdropping on users' private information through USB power cables is possible (Cronin et al., 2021; La Cour et al., 2021; Liu et al., 2022; Spolaor et al., 2023). However, in real-world scenarios, these methods require modifying charging piles to insert attacking devices. Unlike these approaches, our work eliminates the need to conceal additional hardware components in the victim's device, charger, or USB cable. All that is needed is a USB cable to connect the attacking device.

7. Conclusion

This paper introduces a new side-channel attack called Surf-Snooping. This attack exploits the wireless charging process of devices utilizing multi-port USB power supplies. By capturing voltage dynamics from the USB port adjacent to the charging port, the attacker can obtain sensitive information such as the device model, lock screen password, launched applications, and keystrokes. Our work demonstrates the resilience of Surf-Snooping against real-world influencing factors. We aim to raise public awareness regarding the vulnerabilities in USB circuit design as a power interface, enhance security consciousness during wireless charging, and stimulate further research into emerging attacks and defense mechanisms for wireless charging systems.

CRediT authorship contribution statement

Yue Hou: Writing – original draft, Visualization, Data curation. **Xinyan Zhou:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Huakang Xia:** Validation, Supervision, Resources, Formal analysis, Conceptualization. **Jian Wang:** Validation, Supervision, Software, Formal analysis, Conceptualization. **Haiming Chen:** Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

Ba, Z., Zheng, T., Zhang, X., Qin, Z., Li, B., Liu, X., Ren, K., 2020. Learning-based practical Smartphone Eavesdropping with Built-in Accelerometer. In: Network and Distributed Systems Security (NDSS) Symposium 2020, Vol. 2020. pp. 1–18. http://dx.doi.org/10.14722/ndss.2020.24076.

Brouwers, N., Zuniga, M., Langendoen, K., 2014. Neat: A novel energy analysis toolkit for free-roaming smartphones. In: Proceedings of the 12th ACM Conference on Embedded Network Sensor Systems. pp. 16–30. http://dx.doi.org/10.1145/2668332.2668337.

Carroll, A., Heiser, G., 2010. An analysis of Power Consumption in a Smartphone. In: 2010 USENIX Annual Technical Conference (USENIX ATC 10). URL https://www.usenix.org/legacy/event/atc10/tech/full_papers/Carroll.pdf.

Chen, Y., Jin, X., Sun, J., Zhang, R., Zhang, Y., 2017. Powerful: Mobile app fingerprinting via power analysis. In: IEEE INFOCOM 2017-IEEE Conference on Computer Communications. IEEE, pp. 1–9. http://dx.doi.org/10.1109/INFOCOM.2017.8057232.

Cronin, P., Gao, X., Yang, C., Wang, H., 2021. Charger-Surfing: Exploiting a Power Line Side-Channel for Smartphone Information Leakage. In: 30th USENIX Security Symposium (USENIX Security 21). pp. 681–698, URL https://www.usenix.org/conference/usenixsecurity21/presentation/cronin.

Griscioli, F., Pizzonia, M., Sacchetti, M., 2016. USBCheckIn: Preventing BadUSB attacks by forcing human-device interaction. In: 2016 14th Annual Conference on Privacy, Security and Trust. PST, IEEE, pp. 493–496. http://dx.doi.org/10.1109/PST.2016.7907004.

Gurgens, S., Rudolph, C., Mana, A., Munoz, A., 2007. Facilitating the Use of TPM Technologies through S & D Patterns. In: 18th International Workshop on Database and Expert Systems Applications (DEXA 2007). IEEE, pp. 765–769. http://dx.doi.org/10.1109/DEXA.2007.113.

La Cour, A.S., Afridi, K.K., Suh, G.E., 2021. Wireless Charging Power Side-Channel Attacks. In: Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security. pp. 651–665. http://dx.doi.org/10.1145/3460120.3484733.

Li, J., Zhou, H., Wu, S., Luo, X., Wang, T., Zhan, X., Ma, X., 2022. FOAP: Fine-grained open-world android app fingerprinting. In: 31st USENIX Security Symposium (USENIX Security 22). pp. 1579–1596, URL https://www.usenix.org/conference/usenixsecurity22/presentation/li-jianfeng.

Liu, J., Zou, X., Zhao, L., Tao, Y., Hu, S., Han, J., Ren, K., 2022. IEEE transactions on dependable and secure computing. Priv. Leakage Wirel. Charg. http://dx.doi.org/10.1109/TDSC.2022.3173063.

Lu, X., Wang, P., Niyato, D., Kim, D.I., Han, Z., 2016. Wireless charging technologies: Fundamentals, standards, and network applications. IEEE Commun. Surv. & Tutorials 18, 1413–1452. http://dx.doi.org/10.1109/COMST.2015.2499783.

Ma, X., Huang, P., Jin, X., Wang, P., Park, S., Shen, D., Zhou, Y., Saul, L.K., Voelker, G.M., 2013. eDoctor: Automatically diagnosing abnormal battery drain issues on smartphones. In: 10th USENIX Symposium on Networked Systems Design and Implementation (NSDI 13). pp. 57–70, URL https://www.usenix.org/conference/nsdi13/technical-sessions/presentation/ma.

Michalevsky, Y., Schulman, A., Veerapandian, G.A., Boneh, D., Nakibly, G., 2015. PowerSpy: Location tracking using mobile device power analysis. In: 24th USENIX Security Symposium (USENIX Security 15). pp. 785–800, URL https://www.usenix.org/conference/usenixsecurity15/technical-sessions/presentation/michalevsky.

Monaco, J.V., 2018. Sok: Keylogging side channels. In: 2018 IEEE Symposium on Security and Privacy (SP). IEEE, pp. 211–228. http://dx.doi.org/10.1109/SP.2018.00026.

Muñoz A. Rios, R., Román, R., López, J., A survey on the (in) security of trusted execution environments. Comput. Secur. 129, 103180. http://dx.doi.org/10.1016/j.cose.2023.103180.

Neugschwandtner, M., Beitler, A., Kurmus, A., 2016. A transparent defense against USB eavesdropping attacks. In: Proceedings of the 9th European Workshop on System Security. pp. 1–6. http://dx.doi.org/10.1145/2905760.2905765.

Ni, T., Lan, G., Wang, J., Zhao, Q., Xu, W., 2023a. Eavesdropping Mobile App Activity via Radio-Frequency Energy Harvesting. In: 32nd USENIX Security Symposium (USENIX Security 23). USENIX Association, Anaheim, CA, pp. 3511–3528, URL https://www.usenix.org/conference/usenixsecurity23/presentation/ni.

Ni, T., Zhang, X., Zuo, C., Li, J., Yan, Z., Wang, W., Xu, W., Luo, X., Zhao, Q., 2023b. Uncovering user interactions on smartphones via contactless wireless charging side channels. IEEE, pp. 3399–3415. http://dx.doi.org/10.1109/SP46215.2023.10179322.

Owusu, E., Han, J., Das, S., Perrig, A., Zhang, J., 2012. Accessory: password inference using accelerometers on smartphones. In: Proceedings of the Twelfth Workshop on Mobile Computing Systems & Applications. pp. 1–6. http://dx.doi.org/10.1145/2162081.2162095.

Qi-Wireless-Charging.net, 2020. Wireless charging Receiver & Cases: How to Upgrade Smartphones. URL https://qi-wireless-charging.net/wireless-charging-adapter-cases-upgrade-smartphones-with-qi-standard/.

Spolaor, R., Abudahi, L., Moonsamy, V., Conti, M., Poovendran, R., 2017. No free charge theorem: A covert channel via USB charging cable on mobile devices. In: Applied Cryptography and Network Security: 15th International Conference, ACNS 2017, Kanazawa, Japan, July (2017) 10-12, Proceedings 15. Springer, pp. 83–102. http://dx.doi.org/10.1007/978-3-319-61204-1_5.

Spolaor, R., Liu, H., Turrin, F., Conti, M., Cheng, X., 2023. Plug and Power: Fingerprinting USB Powered Peripherals via Power Side-channel. In: IEEE INFOCOM 2023-IEEE Conference on Computer Communications. IEEE, pp. 1–10. http://dx.doi.org/10.1109/INFOCOM53939.2023.10229048.

Su, Y., Genkin, D., Ranasinghe, D., Yarom, Y., 2017. USB Snooping Made Easy: Crosstalk Leakage Attacks on USB Hubs. In: 26th USENIX Security Symposium (USENIX Security 17). pp. 1145–1161, URL https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/su.

Tian, D.J., Bates, A., Butler, K., 2015. Defending Against Malicious USB Firmware with GoodUSB. In: Proceedings of the 31st Annual Computer Security Applications Conference. pp. 261–270. http://dx.doi.org/10.1145/2818000.2818040.

Tian, D.J., Scaife, N., Bates, A., Butler, K., Traynor, P., 2016. Making USB great again with USBFILTER. In: 25th USENIX Security Symposium (USENIX Security 16). pp. 415–430, URL https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/tian.

Tian, J., Scaife, N., Kumar, D., Bailey, M., Bates, A., Butler, K., 2018. Sok: Plug&Pray Today – Understanding USB Insecurity in Versions 1 Through C. In: 2018 IEEE Symposium on Security and Privacy. SP, IEEE, pp. 1032–1047. http://dx.doi.org/10.1109/SP.2018.00037.

Van Wageningen, D., Staring, T., 2010. The Qi wireless power standard. In: Proceedings of 14th International Power Electronics and Motion Control Conference EPE-PEMC 2010. IEEE, pp. S15–25. http://dx.doi.org/10.1109/EPEPEMC.2010.5606673.

Wang, Y., Guo, H., Yan, Q., 2022. GhostTalk: Interactive Attack on Smartphone Voice System Through Power Line. http://dx.doi.org/10.14722/ndss.2022.24254.

Wu, Y., Li, Z., Van Nostrand, N., Liu, J., 2021. Time to rethink the design of Qi standard? Security and privacy vulnerability analysis of Qi Wireless Charging. In: Proceedings of the 37th Annual Computer Security Applications Conference. pp. 916–929. http://dx.doi.org/10.1145/3485832.3485839.

Yan, L., Guo, Y., Chen, X., Mei, H., 2015. A study on power side channels on mobile devices. In: Proceedings of the 7th Asia-Pacific Symposium on Internetware Internetware '15. Association for Computing Machinery., New York, NY, USA, pp. 30–38. http://dx.doi.org/10.1145/2875913.2875934.

Yang, Q., Gasti, P., Zhou, G., Farajidavar, A., Balagani, K.S., 2016. On inferring browsing activity on smartphones via USB power analysis side-channel. IEEE Trans. Inf. Forensics Secur. 1056–1066. http://dx.doi.org/10.1109/TIFS.2016.2639446.

Zhou, M., Wang, Q., Yang, J., Li, Q., Xiao, F., Wang, Z., Chen, X., 2018. Patternlistener: cracking android pattern lock using acoustic signals. In: Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security. pp. 1775–1787. http://dx.doi.org/10.1145/3243734.3243777.

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